

# Transport Geometry Fusion Descent: a sub-second implementation of Meyer’s $G$ -norm cartoon + texture decomposition

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July 21, 2026

## Abstract

Gilles and Osher solve Meyer’s cartoon + texture decomposition by an outer alternation whose inner problems are Rudin–Osher–Fatemi (ROF) solves, implemented with Split Bregman iterations. We show that their Algorithm 3 is exactly a two-projector alternation between a total-variation smoothing and the projection onto Meyer’s  $G$ -ball. We then present transport geometry fusion descent (TGFD), which replaces the nested outer loop with interleaved single Bregman sweeps against persistent auxiliary states. On the  $512^2$  Barbara image used by Gilles and Osher, TGFD reproduces the decomposition reached by the nested algorithm in 361 seconds to a relative distance of  $10^{-2}$ , without visible difference, in 0.9 seconds. We establish three quantitative laws governing acceleration in this problem. First, a momentum polarity: heavy-ball extrapolation of the primal iterates diverges for  $\beta \geq 0.8$ , whereas the same impulse applied to the dual flux — the field  $p$  in Meyer’s definition  $v = \operatorname{div} p$  — accelerates Chambolle’s projector by a factor of 40 at equal iteration count. Second, a scope law for warm starts: Bregman auxiliary fields transfer across outer passes of one subproblem but not across fidelity constants. Third, a soft-rung law for scale decompositions on the  $G$ -axis: the capture fraction of an oscillation under the  $G$ -ball projection is a linear ramp of 10–90% width  $9\text{--}14\times$  in  $\mu$ , so that ROF scale ladders are graded rather than cellular, admit no critical rung ratio, and support approximately two separable texture bands per image. A three-rung ladder nevertheless isolates the contour-leakage artifact of the single- $\mu$  decomposition into its coarsest band, yielding a textile band that no single- $\mu$  decomposition produces. We conclude with an account of the measured principles, obtained in a precursor program on time-frequency super-resolution, from which the method derives.

## 1 Introduction

The cartoon + texture model of Meyer [1] decomposes an image  $f$  into a bounded-variation component  $u$  and an oscillatory component  $v$  constrained to the ball  $G_\mu = \{v : \|v\|_G \leq \mu\}$  of the  $G$ -norm,

$$\|v\|_G = \inf\{\|p\|_\infty : v = \operatorname{div} p\}, \quad (1)$$

the infimum taken over vector fields  $p$  whose divergence generates  $v$ . Aujol, Aubert, Blanc-Féraud and Chambolle [2] made the model computable by alternating two subproblems, each solved with Chambolle’s projection algorithm [3]. Gilles and Osher [5] replaced the inner projector with Split Bregman ROF solves [4] and observed a substantial reduction in inner iterations.

The outer loop of [5] — initialization at  $u_0 = v_0 = 0$ , inner solves to tolerance, alternation to convergence — is inherited unchanged from [2].

We use the model, the inner solver, the test image, and the parameter roles of [5], and we revise the outer computation. The contributions are the following.

1. A reduction (Proposition 3.1): Algorithm 3 of [5] is exactly the alternation  $u \leftarrow \text{ROF}(f - v, \lambda)$ ,  $v \leftarrow P_{G_\mu}(f - u)$ .
2. A method (Algorithm 1): transport geometry fusion descent, an interleaved single-sweep form of that alternation. On the rig of Section 5 it reaches a lower fixed-point residual than the nested algorithm at  $26\times$  fewer inner sweeps; on the  $512^2$  Barbara image it reproduces the nested decomposition to  $\|\Delta u\| / \|f\| = 1.0 \times 10^{-2}$  at a  $\sim 400\times$  reduction in wall time.
3. A momentum polarity law (Section 7): primal heavy-ball extrapolation diverges, while FISTA extrapolation of the dual flux accelerates the texture projection by a factor of 40; momentum carried across a changing objective is harmful on both sides of the duality gap.
4. A soft-rung law (Section 8): the  $G$ -ball capture curve of an oscillation is a linear ramp, with the consequences that two-way band routing depends only on the straddling rung, that optimal rung placement is skewed toward the finer texture, and that the  $G$ -axis supports approximately two separable texture bands per image.
5. A band decomposition of the Barbara texture layer (Section 9) in which the contour-leakage artifact of the single- $\mu$  model concentrates in the coarsest band.
6. An account of the measured principles from which the method derives (Section 10), stated so that the present paper is self-contained.

All quantitative claims are measurements obtained under the protocol of Section 5; the mathematics required to reproduce them is contained in Sections 2–5.

## 2 Preliminaries and discretization

Images are arrays in  $[0, 255]^{N \times M}$ , treated as real-valued with periodic boundary conditions. The discrete gradient uses forward differences,

$$(\nabla u)_{i,j} = (u_{i,j+1} - u_{i,j}, u_{i+1,j} - u_{i,j}),$$

indices modulo the array dimensions; the divergence is its negative adjoint,  $(\text{div } p)_{i,j} = p_{i,j}^x - p_{i,j-1}^x + p_{i,j}^y - p_{i-1,j}^y$ . The total variation is the isotropic sum  $J(u) = \sum_{i,j} |(\nabla u)_{i,j}|$ . The composition  $\Delta = \text{div } \nabla$  is diagonalized by the discrete Fourier transform with symbol

$$\hat{\Delta}(k, l) = 2 \cos(2\pi k/N) - 2 + 2 \cos(2\pi l/M) - 2 \leq 0. \quad (2)$$

**ROF and the  $G$ -ball projection.** For a fidelity coefficient  $c > 0$  we write

$$\text{ROF}(g, c) = \arg \min_w J(w) + \frac{c}{2} \|g - w\|_2^2.$$

Chambolle’s identity [3] states that the ROF residual is the projection onto the  $G$ -ball of radius  $1/c$ :

$$g - \text{ROF}(g, c) = P_{G_{1/c}}(g). \quad (3)$$

**The Split Bregman sweep.** Split Bregman [4] solves  $\text{ROF}(g, c)$  by introducing  $d \approx \nabla u$  with penalty  $\eta$  and Bregman field  $b$ . One sweep against the state  $(d, b)$  consists of

$$u \leftarrow \mathcal{F}^{-1} \left[ \frac{c \hat{g} - \eta \widehat{\text{div}(d - b)}}{c - \eta \hat{\Delta}} \right], \quad (4)$$

$$t = \nabla u + b, \quad d \leftarrow \max(|t| - \frac{1}{\eta}, 0) \frac{t}{|t|}, \quad (5)$$

$$b \leftarrow b + \nabla u - d, \quad (6)$$

where  $\mathcal{F}$  denotes the FFT, the shrinkage (5) is isotropic in the two gradient components, and the linear solve (4) is exact by (2). One sweep is the unit of computational cost throughout this paper.

**The dual flux and fast gradient projection.** The dual of  $\text{ROF}(g, c)$  is a constrained least-squares problem in the flux variable of (1):

$$p^* = \arg \min_{|p| \leq 1} \|g + \text{div } p/c\|_2^2, \quad u = g + \text{div } p^*/c. \quad (7)$$

Projected gradient descent on (7) with step  $c/8$  is Chambolle’s algorithm; adding FISTA extrapolation of  $p$  yields fast gradient projection (FGP) [6]: with  $t_0 = 1$ ,

$$\begin{aligned} y_k &= p_k + \frac{t_{k-1}-1}{t_k} (p_k - p_{k-1}), & u_k &= g + \text{div } y_k/c, \\ p_{k+1} &= \Pi(y_k + \frac{c}{8} \nabla u_k), & t_{k+1} &= \frac{1 + \sqrt{1 + 4t_k^2}}{2}, \end{aligned} \quad (8)$$

where  $\Pi$  is the pointwise projection onto the isotropic unit ball.

### 3 Reduction of the Gilles–Osher algorithm

In [5] the operator  $\text{P}_{\text{ROF}}(g, c)$  denotes the ROF solution with fidelity coefficient  $c$ ; this convention is fixed by the proof of their Proposition 2. Their Algorithm 3 iterates

$$u_{n+1} = \text{P}_{\text{ROF}}(f - v_n, \lambda), \quad v_{n+1} = f - u_{n+1} - \frac{1}{\lambda} \text{P}_{\text{ROF}}(\lambda(f - u_{n+1}), \frac{1}{\lambda\mu}).$$

**Proposition 3.1.** *Algorithm 3 of [5] is identical to the two-projector alternation*

$$u_{n+1} = \text{ROF}(f - v_n, \lambda), \quad v_{n+1} = P_{G_\mu}(f - u_{n+1}). \quad (9)$$

*Proof.* Substituting  $w = \lambda z$  in the objective of the texture step gives  $J(w) + \frac{1}{2\lambda\mu} \|\lambda g - w\|_2^2 = \lambda [J(z) + \frac{1}{2\mu} \|g - z\|_2^2]$ , whence  $\frac{1}{\lambda} \text{P}_{\text{ROF}}(\lambda g, \frac{1}{\lambda\mu}) = \text{ROF}(g, \frac{1}{\mu})$ . The texture update is therefore  $g - \text{ROF}(g, \frac{1}{\mu}) = P_{G_\mu}(g)$  by (3).  $\square$

Equation (9) exposes the transfer mechanics of the outer loop. The cartoon step removes from  $u$  a component of  $G$ -norm at most  $1/\lambda$ ; the texture step assigns to  $v$  whatever portion of  $f - u$  lies within the  $\mu$ -ball. A clean decomposition requires  $1/\lambda$  well below the  $G$ -norm of the texture, and under that condition the outer loop is a geometric transfer of texture mass from  $u$  to  $v$  at a rate limited by the sliver size  $1/\lambda$ . The alternation is therefore an instance of constraint-family fusion on a single latent image, a structure whose computational behavior we had previously characterized in the setting described in Section 10. The principles measured there transfer to (9), with polarities that the convexity of the present problem reverses in specific and instructive ways.

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**Algorithm 1** Transport geometry fusion descent (TGFD)

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1:  $u \leftarrow 0, v \leftarrow 0$ ; Bregman states  $(d_u, b_u) \leftarrow 0, (d_v, b_v) \leftarrow 0$ 
2: while the wall-time budget remains do
3:   apply one sweep (4)–(6) for  $\text{ROF}(f - v, \lambda)$ ,  $\eta = 2\lambda$ , against  $(d_u, b_u)$ ; obtain  $u$ 
4:   apply one sweep (4)–(6) for  $\text{ROF}(f - u, 1/\mu)$ ,  $\eta = 10/\mu$ , against  $(d_v, b_v)$ ; obtain  $w$ 
5:    $v \leftarrow (f - u) - w$ 
6: end while
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## 4 Transport geometry fusion descent

The method is stated in Algorithm 1. Each outer pass performs one Split Bregman sweep on each subproblem of (9), against auxiliary fields that persist for the life of the loop. There is no inner tolerance, no nesting, no momentum, no seed, and no continuation schedule. The texture-side penalty is  $\eta = 10/\mu$  rather than the customary  $2c$ ; at weak fidelity this value is the accuracy-per-sweep optimum among the settings we tested (Table 2).

Each component absent from Algorithm 1 was implemented, measured, and removed on the evidence.

- *Nesting.* Inner solves to tolerance expend sweeps at outer states that are still far from the fixed point. Removing the nesting reduces cost by a factor of approximately 26 at equal or better fixed-point residual (Table 1).
- *Primal momentum.* Heavy-ball extrapolation of  $(u, v)$  is neutral for  $\beta \leq 0.6$  and divergent beyond  $\beta \approx 0.8$  (Section 7).
- *Seeding.* A separate initialization pass is redundant: under single-sweep interleaving from zero, the early passes execute against a coarse auxiliary state and perform the function of a seed. Explicit seeds of 5 and 40 sweeps produced no measurable improvement.
- *Over-relaxation, continuation, multiresolution.* Krasnosel’skiĭ–Mann relaxation with  $\omega \in [1.3, 1.9]$ , geometric  $\lambda$ -continuation from  $\lambda/4$  through  $\lambda/16$ , and a half-resolution initialization were each tested at fixed wall budget; all lie within measurement noise of Algorithm 1 (Section 6), and the half-resolution variant is measurably worse on Barbara, where down-sampling destroys texture near the Nyquist frequency before the full-resolution passes can restore it.

## 5 Experimental protocol

**Synthetic scene.** The controlled experiments use a  $256^2$  scene with known ground truth on the grid  $(x, y) \in \{0, \dots, n-1\}^2$ ,  $n = 256$ . The cartoon component is the affine field  $90 + 40x/n + 20y/n$ , overwritten by a disk of center  $(0.32n, 0.34n)$  and radius  $0.18n$  carrying the value  $170 - 30y/n$ , and by the rectangle  $[0.55n, 0.9n] \times [0.55n, 0.85n]$  carrying the value 60. Two texture components are added, each of the form  $a m(x, y) \cos(2\pi(x \cos \theta + y \sin \theta)/T)$  with amplitude  $a = 20$  and window  $m = \max(0, 1 - \rho)^{1/2}$ ,  $\rho$  the distance to the window center in units of  $0.3n$ : a coarse texture with  $T = 12$ ,  $\theta = 0.4$ , centered at  $(0.38, 0.62)n$ , and a fine texture with  $T = 3$ ,  $\theta = 1.2$ , centered at  $(0.62, 0.40)n$ . The  $G$ -norms of the textures are approximately  $aT/2\pi = 38.2$  and  $9.5$ . Unless stated otherwise,  $\lambda = 0.05$  and  $\mu = 60$  on the synthetic scene, and  $\lambda = 0.05$ ,  $\mu = 40$  on Barbara ( $512^2$ , 8-bit grayscale, values scaled to  $[0, 255]$ ).

Table 1: Synthetic scene,  $\lambda = 0.05$ ,  $\mu = 60$ .

| variant                                   | sweeps | wall   | $r(u, v)$            | PSNR $u$ | PSNR $v$ |
|-------------------------------------------|--------|--------|----------------------|----------|----------|
| Algorithm 3 of [5], nested, tol $10^{-4}$ | 21,071 | 18.7 s | $2.4 \times 10^{-3}$ | 35.01    | 36.21    |
| TGFD, 800 passes                          | 800    | 0.96 s | $2.0 \times 10^{-3}$ | 34.55    | 35.66    |
| TGFD + primal heavy-ball $\beta = 0.88$   | 800    | 0.85 s | $5.9 \times 10^{-2}$ | 27.86    | 24.52    |

**Reference fixed point.** Errors are measured against a reference computed by 120 exact outer alternations of (9), each subproblem solved by 2000 FGP iterations (8). The residual drift of this reference over its final 60 alternations is  $2.0 \times 10^{-3}$  relative to  $\|f\|$ , which is therefore the resolution floor of every comparison below.

**Metrics.** The fixed-point residual of a pair  $(u, v)$  is

$$r(u, v) = \max(\|u - \text{ROF}(f - v, \lambda)\|_2, \|v - P_{G_\mu}(f - u)\|_2) / \|f\|_2,$$

with the two operators evaluated by high-accuracy solves. PSNR is computed with peak 255. Cost is reported in sweeps and in single-core NumPy seconds; no compiled code and no GPU are used.

## 6 Results

Table 1 contains the central comparison. TGFD reaches a lower fixed-point residual than the nested algorithm at  $26\times$  fewer sweeps, and converges to the same decomposition: the relative distance  $\|u_{\text{TGFD}} - u_{\text{nested}}\| / \|f\|$  is  $9 \times 10^{-3}$ , within a factor of five of the reference drift floor. The third row demonstrates that the single mechanism added to the alternation — primal momentum — is the mechanism that degrades it.

**Measured Result 6.1** (Barbara,  $512^2$ ,  $\lambda = 0.05$ ,  $\mu = 40$ ). Nested Algorithm 3 at tolerances  $10^{-5}$  requires 63,633 sweeps and 361 s. TGFD at a 0.9-second budget completes 60 passes and reaches  $\|u - u_{\text{nested}}\| / \|f\| = 1.0 \times 10^{-2}$ , a  $\sim 400\times$  reduction in wall time; a 2.5-second budget reaches  $5.8 \times 10^{-3}$ . The difference of the two cartoons is an edge-adjacent ripple visible only under  $20\times$  amplification (Figure 1). For context, a single texture-side projection computed with Chambolle’s algorithm at tolerance  $10^{-4}$  requires 4000 iterations of (7); the inner-solver comparison of [5] is thereby reproduced in passing.

A budget study closes the question of further per-pass acceleration at this scale. With wall time capped at 0.8 s on the synthetic scene, eleven variants — penalty retuning, over-relaxation, continuation, seeding, and their combinations, as listed in Section 4 — produce errors confined to  $[0.9, 1.6] \times 10^{-2}$  with PSNR 34.1–34.5 dB. The decomposition is PSNR-converged by approximately 130 passes; the residual error beyond that point is outer-transfer distance, and no per-pass mechanism among those tested removes it. We therefore regard Algorithm 1, without additions, as the operating point of this problem class at sub-second budgets.

## 7 Momentum and the dual flux

Meyer’s  $G$ -norm (1) is a transport norm: the texture is the divergence of a flux field, and the norm is the least uniform bound on that flux. The dual problem (7) shows that the ROF



Figure 1: Barbara. Top row: input; cartoon and texture computed by nested Algorithm 3 (361 s). Bottom row: TGFD at a 0.9-second budget; difference to the nested cartoon, amplified  $20\times$ .

subproblems of (9) are themselves optimizations over the same flux. The name of the method records this geometry: the quantities that move mass between the layers of the decomposition, and the quantities that accept acceleration, are the fluxes rather than the images.

**Measured Result 7.1** (Momentum polarity). Heavy-ball extrapolation of the primal iterates of (9) is neutral for  $\beta \leq 0.6$  and divergent for  $\beta \geq 0.8$ : the fixed-point residual moves from  $2.0 \times 10^{-3}$  at  $\beta = 0$  to  $2.7 \times 10^{-2}$  at  $\beta = 0.8$  and  $5.9 \times 10^{-2}$  at  $\beta = 0.88$ . The same impulse applied to the dual flux — FISTA extrapolation of  $p$  in (8) — reduces the error of the texture projection by a factor of 40 at equal iteration count relative to the identical iteration without extrapolation (Table 2).

**Measured Result 7.2** (Stale transport momentum). Within the full alternation, where each pass changes the subproblem objective, carrying the FISTA extrapolation state across passes yields pipeline error  $7.6 \times 10^{-2}$ ; carrying the flux while restarting the extrapolation sequence each pass yields  $1.9 \times 10^{-2}$ . Momentum constructed for one transport flow misdirects the next. Together with Measured Result 7.1, this identifies a single phenomenon appearing on both sides of the duality gap.

The same scope law governs the warm state itself. The Bregman fields  $(d, b)$  transfer across outer passes of one subproblem; they do not transfer across fidelity constants, being  $\eta$ - and



Table 2: Texture projection at weak fidelity ( $c = 1/60$ ): relative error to the converged projection after 800 inner iterations, and per-iteration cost.

| solver                                              | error @ 800 it       | ms/it |
|-----------------------------------------------------|----------------------|-------|
| Chambolle iteration on (7)                          | $1.9 \times 10^{-1}$ | 0.31  |
| FGP (8): the same iteration with flux extrapolation | $5.1 \times 10^{-3}$ | 0.46  |
| Split Bregman, $\eta = 2c$                          | $3.1 \times 10^{-2}$ | 0.82  |
| Split Bregman, $\eta = 10c$                         | $4.0 \times 10^{-3}$ | 0.80  |

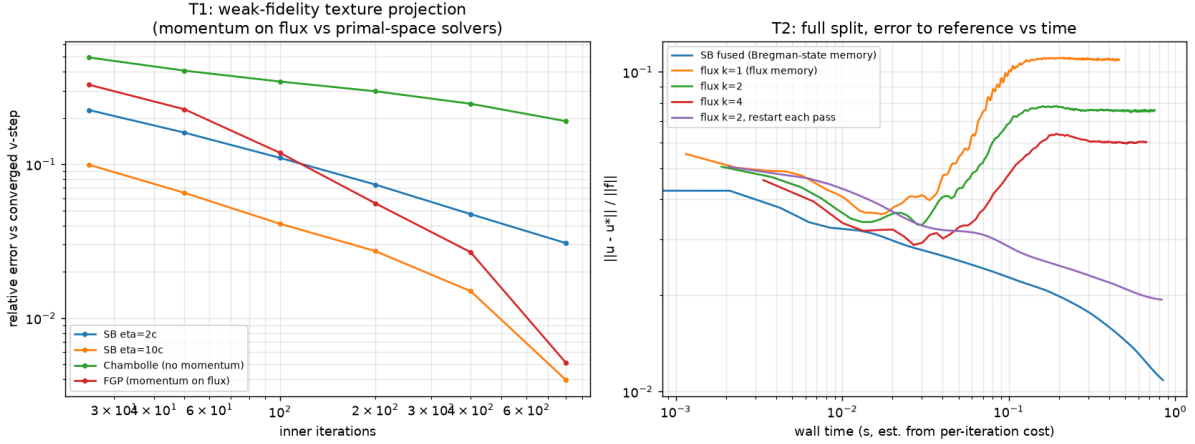


Figure 2: Left: the weak-fidelity texture projection under four solvers. Right: full-pipeline error to the reference fixed point against wall time.

$c$ -scaled, and a state carried to a different constant pins the new solve at the old fixed point. Within the full pipeline at equal wall time, the Split Bregman interleave retains the advantage over pure flux iteration ( $1.1 \times 10^{-2}$  against  $1.9 \times 10^{-2}$ ): the FFT step (4) solves its linear subproblem exactly once per sweep, and the  $(d, b)$  state carries the edge set of the solution across passes. Flux acceleration is decisive where a projection must be computed accurately in isolation — within Chambolle’s projector it should always be used — while the pipeline favors the warm interleave.

## 8 The scale geometry of the $G$ -axis

A natural extension of the decomposition is a scale ladder: rungs  $\mu_0 > \mu_1 > \dots > \mu_K$  and texture bands defined as differences of the ROF scale space,

$$s_k = \text{ROF}(v, 1/\mu_k), \quad B_k = s_{k+1} - s_k, \quad B_K = v - s_K, \quad (10)$$

so that  $v = s_0 + \sum_k B_k$ . Before constructing ladders we measured the geometry of the axis they inhabit.

**Measured Result 8.1** (The capture curve is a ramp). For a pure oscillation  $g$  of  $G$ -norm  $G$ , the captured fraction  $\|P_{G_\mu}(g)\|_2 / \|g\|_2$  as a function of  $\mu/G$  has 10–90% transition width  $e^{2.2}$  to  $e^{2.6}$ , i.e. 9–14 $\times$  in  $\mu$ , for periods 4, 8, and 16 pixels alike (Figure 3).

This is the width that theory assigns: ROF acts on an oscillation by soft-thresholding its amplitude, so the captured fraction is approximately  $\min(1, \mu/G)$  — a linear ramp, whose 10–90 width is  $9\times$  exactly. Three consequences follow, each verified on the two-texture scene of Section 5.

**Lemma 8.2** (Routing telescopes). *Let bands be defined by (10) and let a two-way separation assign each band to one of two layers by a threshold on its rung interval. Then the layers depend only on the scale-space values at the two rungs adjacent to the threshold; every interior rung cancels.*

*Proof.* The sum of consecutive bands telescopes:  $\sum_{k=j}^{m-1} B_k = s_m - s_j$ .  $\square$

Indeed, ladders of ratio 2 and ratio 4 over the same span produce bitwise-identical two-way layers in our measurements. Consequently the design of a two-way separation reduces to the placement of a single straddling rung; and because the capture curve is a wide ramp rather than a step, the optimal placement is not the geometric midpoint of the two texture scales. On the two-texture scene, a rung at  $0.79 G_{\text{fine}}$  yields total cross-leakage 0.20/0.37 (coarse layer evaluated in the fine texture’s region and conversely), against 0.14/0.65 for the midpoint rung: the skewed placement is superior because the coarse texture’s ramp value at a low rung is smaller than the fine texture’s ramp value at the midpoint.

The axis is, moreover, shallow. Cross-talk between bands decays only linearly in scale separation, so a clean separation requires a pair ratio comparable to the transition width, approximately  $14\times$ ; and the usable span is bounded above by collision with the cartoon scale. When the coarse texture’s period was raised to 48 pixels ( $G = 153$ ) and  $\mu$  raised accordingly, the  $u/v$  decomposition itself degraded by 6.8 dB: the  $G$ -ball at that radius absorbs cartoon structure. Between the ramp below and the cartoon above, the  $G$ -axis of a  $256^2$ – $512^2$  image supports approximately two genuinely separable texture bands.

We record that the opposite result was expected. In the precursor program (Section 10) we had established a sharp-cell ladder theorem for windowed-spectral constraint families: geometric ladders of scale ratio 4 cover their span without interior holes, ratio 8 collapses, and a critical ratio therefore exists in  $(4, 8]$ . The present measurements locate the boundary of that theorem’s validity. Sharp cells are a property of the analyzing family, not of multiscale decomposition as such; on the  $G$ -axis there is no cell, no hole, and no critical ratio. Sharp multi-band texture routing therefore belongs to windowed-spectral families, while the  $G$ -ladder yields graded bands.

## 9 Band separation on Barbara

Figure 2 of [5] exhibits the standing artifact of the single- $\mu$  model: contours of the cartoon — the face line, the table edge — appear in the texture layer, because a single  $G$ -ball cannot distinguish a sharp edge from a high-amplitude oscillation. A graded ladder does not remove this leakage; it sorts it.

**Measured Result 9.1** (Barbara, rungs  $\mu = \{40, 10, 2.5\}$ ). On the converged texture layer, the band energies divide 5365 / 3338 / 1589, with 17 units of coarsest survivor returned to the cartoon. The contour-leakage content concentrates in the coarse band ( $\mu 40 \rightarrow 10$ ); the mid band ( $\mu 10 \rightarrow 2.5$ ) contains the textile content — scarf stripes and tablecloth check — with little contour contamination; the fine band contains weave detail and grain (Figure 4). The cost is three independent ROF solves, 1.8 s at  $512^2$ .



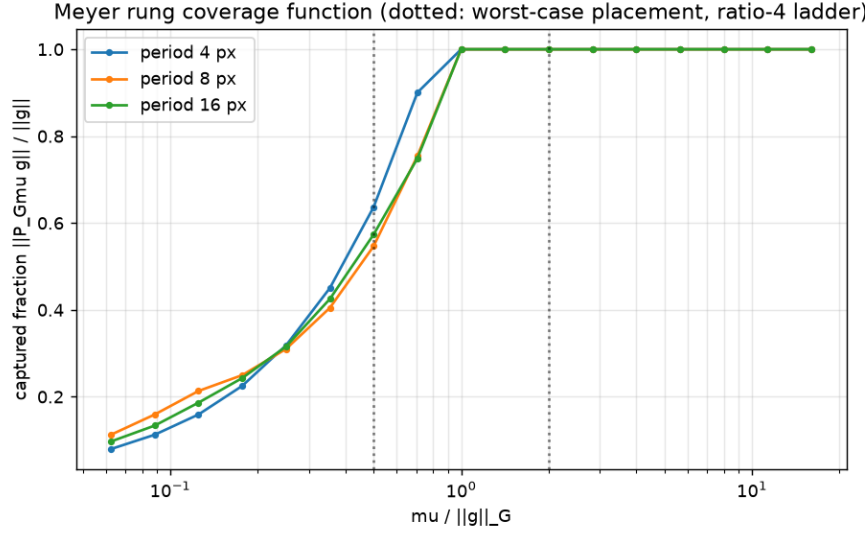


Figure 3: The capture fraction of a pure oscillation under  $P_{G_\mu}$  against  $\mu / \|g\|_G$ , for three periods. Dotted lines mark the worst-case rung placements of a ratio-4 ladder.

A consumer of texture information reads the mid band and receives a layer that the single- $\mu$  decomposition cannot produce at any parameter setting, since its  $v$  mixes the three bands by construction. The bands must be computed as independent solves; this is the warm-state scope law of Section 7 in operation, since a Bregman state carried across rungs pins the next solve at the previous fixed point and empties the corresponding band.

## 10 Origins of the method

The method presented above is an application of principles measured in a precursor program on time-frequency analysis. We give that history in terms of the questions that drove it, the mathematical settings in which each question was posed, and the truths that emerged; the reader needs no familiarity with the program, and the statements below are quantitative. Transport-geometric objects recur throughout it, first on the display side, then as a rejected alternative for fusion, and finally — in the present paper — as the carrier of acceleration; this recurrence, and not any single result, is what the method’s name commemorates.

### 10.1 Fusion as intersection

The program began with a question about spectrogram displays of sampled radio signals: a long analysis window resolves frequency and blurs time, a short window resolves time and blurs frequency; both are computed from the same record; can one produce a display with the strengths of both?

The prevailing answers operate on the displays. Barycentric fusion computes an optimal-transport interpolant between the two spectrograms treated as nonnegative measures [8]: mass is moved between the two pictures under structured transport costs until a compromise picture is reached. When considering this family of methods we observed that the compromise object is

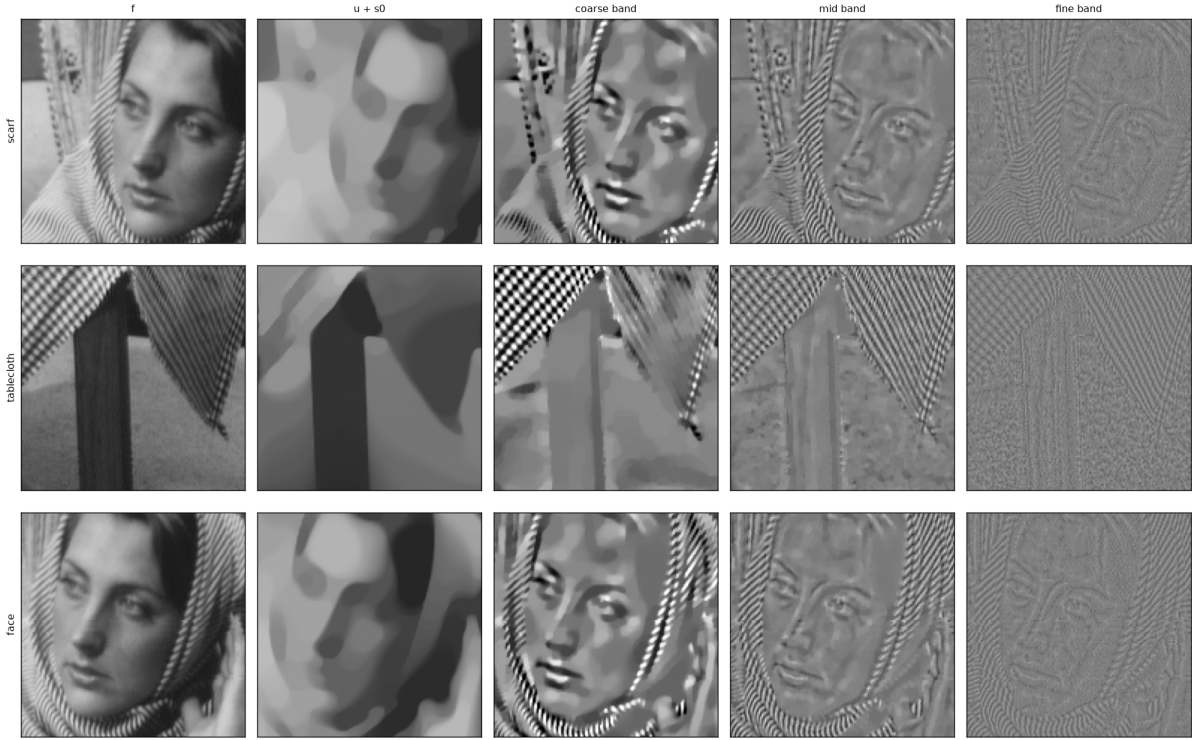


Figure 4: Ladder bands on Barbara in three regions. Columns: input; cartoon plus coarsest survivor; coarse band ( $\mu 40 \rightarrow 10$ ); mid band ( $\mu 10 \rightarrow 2.5$ ); fine band. Contour leakage resides in the coarse band; the mid band carries the textile.

generally not the spectrogram of any signal; the transport prior supplies the joint time-frequency structure that neither display contains.

We instead posed the problem in the signal domain. Let  $A_1, A_2$  be the windowed Fourier analysis operators of the two window lattices, and let  $m_i = |A_i z|$  be the measured magnitudes of the record  $z$ . Each measurement defines a constraint family

$$M_i = \{x : |A_i x| = m_i\},$$

and the natural fusion is an element of  $M_1 \cap M_2$ , sought by alternating the Griffin–Lim projection pair [7]

$$P_i(x) = A_i^+(m_i \cdot A_i x / |A_i x|),$$

where  $A_i^+$  is the least-squares synthesis. Three measurements fixed the program’s direction. First, the record  $z$  is an exact fixed point of the alternation: the residual at  $z$  is zero to machine precision, and therefore fusion of complementary measurements is intersection of constraint families in the latent space rather than compromise between displays. Second, the interference structure carries the joint information: for two tones separated by less than the long window’s resolution, the short-window magnitudes contain the beating term that the long window suppresses, and a controlled rotation experiment showed the short family constrains the pair’s relative phase twice as strongly as the long family. The joint time-frequency content that barycentric methods supply by prior is, in the signal domain, measured data. Third, the iteration’s cost structure is bimodal: from a cold start the alternation stagnates (residual contraction

0.996–0.998 per iteration), while from a physically informed initialization it converges essentially in one step (fidelity  $0.81 \rightarrow 0.93$  in a single iteration, 0.99 by 24). The initialization purchases the basin; the projections purchase joint consistency. Heavy-ball extrapolation at  $\beta = 0.88$  measurably accelerated both regimes of these nonconvex radial projections — the datum that Measured Result 7.1 later inverted in the convex setting.

## 10.2 Coverage governs, operators polish

The same algebra was then applied to images: two windowed-DFT patch families — large patches resolving spatial frequency, small patches resolving localization — with nonnegativity, fusing two degraded captures of one scene. On a scene constructed so that each family resolves what the other cannot, the fusion reached 28.5 dB where either family alone reached 21–23 dB. The question then became: what governs the attainable gain?

A sequence of controlled-blur studies answered it. Two captures through identical blurs yield exactly no gain — the method does not hallucinate. Two captures through complementary blurs (orthogonal motion kernels) yield a reconstruction better than either capture, recovering structure neither contains alone. Upgraded splitting operators — Douglas–Rachford and relaxed variants — yield nothing when the residual error lies in frequency bands that no capture measured, and yield +2.4 dB in a matched problem whose error was of optimization type. Therefore recovery quality tracks the spectral coverage of the capture set, and operator machinery only polishes toward what is covered. The methodological corollary — measure the geometry of the constraint axis before building machinery upon it — dictated the order of operations in Section 8, where the capture curve was measured first and correctly predicted that no operator on the  $G$ -axis routes scales sharply.

## 10.3 Ladders and their cells

The coverage law raised a design question: how many constraint families, at which scales, cover a working range? For windowed-spectral families the answer sharpened into a theorem. The strength with which a window of length  $N$  constrains a component pair separated by  $(\Delta t, \Delta f)$  is a computable spectrogram overlap (log-log correlation 0.9995 against a closed form over three decades of scale), and geometric window ladders of scale ratio 4 cover their span with no interior holes while ratio 8 collapses to a measured zero; a critical ratio therefore exists in  $(4, 8]$ , and the number of rungs required for a scale span is logarithmic. Convergence rate obeys the same geometry: content deep within a rung’s cell converges in tens of iterations, content at a cell edge in hundreds, with rate approximately the square of the pin strength.

These results supplied the expectations that Section 8 tested on Meyer’s axis — and the value of carrying a quantitative theorem across domains is precisely that its failure is informative. The  $G$ -ball capture curve is a ramp, not a cell; the critical-ratio phenomenon is a property of windowed-spectral analysis, not of multiscale decomposition; and the correct design rule on the  $G$ -axis (a single skewed straddling rung) is different in kind from the correct rule on the spectral axis (a logarithmic ladder at ratio 4).

## 10.4 The present work

Proposition 3.1 identifies the Gilles–Osher algorithm as a member of the class the program had characterized: an alternation of projections onto two measured constraint families of one latent image. The transfer of each principle then produced the results above. The interleaving law

transferred directly and yielded the  $400\times$  of Measured Result 6.1. The momentum law transferred with reversed polarity: the  $\beta = 0.88$  gain of the nonconvex setting became the divergence of Table 1, and the question of where the impulse belongs was settled by the problem’s own transport structure — Meyer’s norm is a flux norm, the dual variable is the flux, and extrapolation of that flux is sound and fast (Measured Result 7.1). The coverage-before-machinery rule dictated measuring the capture curve before building ladders, which prevented a class of constructions that could not have worked and produced the band separation that does (Measured Result 9.1). Each transfer was a measurement; where the precursor law held, we used it, and where it failed, the failure localized the law’s domain of validity.

## 11 Conclusion

Algorithm 3 of [5] is a two-projector alternation, and treating it as such reduces its cost by two orders of magnitude: one warm Split Bregman sweep per subproblem per pass reproduces the nested decomposition of the  $512^2$  Barbara image to  $10^{-2}$  in 0.9 seconds. Acceleration in this problem resides in its transport geometry: momentum is sound on the dual flux and unsound on the primal images; auxiliary state transfers along one flow and not across fidelity constants; and the scale structure of Meyer’s axis is a ramp whose geometry admits graded band separation — sufficient to isolate the model’s contour-leakage artifact in its own band — but not sharp multiscale routing. The decomposition, its cost, and the laws governing both are reproducible from the equations and parameters given in Sections 2–5.

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